TECHNICAL DOCUMENTATION V1.0

Mod Alpha

Smarter SIP timing for mutual funds & ETFs — Standard vs Cost-Optimized technical plan

Complete technical requirements for mobile app, AI-powered website, backend, and infrastructure — with two deployment scenarios side by side.

Product	Mod Alpha — SIP timing advisory
Target launch	31 December 2026
Reference product	returns.trade (stocks) → Mod Alpha (SIP/MF/ETF)
Document date	27 September 2026
Scenarios covered	Scenario A: Standard · Scenario B: Cost-Optimized
Scale targets	Year 1: 50k users · Long-term: millions

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1. Executive overview

Mod Alpha connects to an investor's existing broker account, reads MF/ETF holdings, and recommends the best date(s) each month to invest instead of a fixed SIP date — with optional one-tap order placement where the broker API allows.

MF + ETF	1–2 dates	Extra Earnings
In scope at launch	Smart purchases per fund / month	Shadow portfolio vs fixed SIP date

Product surfaces

SURFACE	ROLE	LAUNCH PRIORITY
Flutter mobile app	Primary product — KYC, broker link, smart dates, orders, subscriptions	P0
AI marketing website	Conversion, education, performance proof, compliance — similar positioning to returns.trade but for SIP	P0 (recommended)
Logged-in web app	Full product in browser	Out of scope (PRD) — post-launch
Admin console	Staff 2FA, kill switch, audit logs	P0 (internal)

> **Two scenarios in this document**
>
> **Scenario A (Standard):**
>
> Built for scale from day one — EKS, Multi-AZ, warm DR, multiple broker order APIs, LLM explanations, daily AA refresh.
>
> **Scenario B (Cost-Optimized):**
>
> Lean launch stack — ECS/single-region, Zerodha-only orders, push-first notifications, weekly AA for inactive users, no LLM at v1. Scale when revenue justifies it.

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2. Benchmark: returns.trade → Mod Alpha

returns.trade is a SEBI-registered, NSE-empanelled algo platform for **stocks and F&O**. Mod Alpha applies the same *category* of product — AI/systematic investing on the user's own broker account — to **mutual fund and ETF SIPs**.

RETURNS.TRADE (STOCKS)	MOD ALPHA (SIP / MF / ETF)
Research-backed trade calls (intraday, swing, positional)	Monthly smart-date recommendations (1–2 purchase dates per fund)
Algos execute on user's demat via broker API	One-tap MF/ETF purchase where API supports; else reminder + deep link
Transparent track record & past performance	**Extra Earnings tracker** — shadow portfolio vs user's original SIP date
Money stays in user's broker account	Same — Mod Alpha never holds or moves money
SEBI IA + NSE algo empanelment	SEBI IA registration + broker algo tagging (ETF orders)
Website: trust, research proof, app install CTA	Website: SIP timing education, backtested results, PaRRVA-compliant performance, app install
AI / quant research for equity signals	AI timing engine on NAV/volatility — batch model, fully auditable

> **Website positioning (like returns.trade, adapted for SIP)**
> "No capital lock-ins. No handing over your money. No black boxes. A regulated system operating on your own broker account — but for SIP timing, not stock tips."

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3. Website AI features (SIP-focused)

The Mod Alpha website is not just marketing — it showcases AI capabilities, builds trust with verifiable performance, and drives app installs. Inspired by returns.trade's transparency + automation narrative, adapted for long-term SIP investors.

AI SIP Timing Demo

Interactive widget: user picks a fund + monthly amount → site shows simulated smart dates vs fixed-date SIP over 5 years using historical NAV (clearly labeled "illustration, not advice").

Extra Earnings Preview

Public backtest dashboard: "If you had followed Mod Alpha dates on Nifty 50 Index Fund since 2020..." — PaRRVA-verified where used in marketing.

Smart Schedule Generator

Logged-out flow: enter fund name → AI generates next month's suggested dates (teaser); full schedule + execution requires app + KYC.

AI Explainability Panel

For each recommended date: plain-language rationale (volatility window, mean-reversion score) — optional LLM copy layer on top of deterministic model outputs.

Fund-ready Calendar

Visual monthly calendar with smart dates, fund-ready reminders, and NAV cut-off times — mirrors in-app experience on web for SEO.

Comparison Calculator

Fixed SIP date (e.g. 1st of month) vs Mod Alpha dates — side-by-side unit accumulation chart using historical data.

AI FAQ / Support Bot

RAG chatbot trained on compliance docs, FAQ, and product mechanics — **never** gives personalized investment advice without KYC.

Trust & Compliance Hub

SEBI IA registration, MITC, grievance officer, SCORES link, methodology whitepaper, model version changelog.

Website tech stack

LAYER	SCENARIO A (STANDARD)	SCENARIO B (COST-OPTIMIZED)
Framework	Next.js 15 (SSR + ISR for AI demo pages)	Astro or Next.js SSG only
Hosting	Vercel Pro or CloudFront + Lambda@Edge	S3 + CloudFront (static)
AI demo API	Dedicated NestJS endpoints + Redis cache	Pre-computed JSON bundles refreshed weekly
Charts	Recharts / D3 live from API	Static chart images or client-side from bundled data
FAQ bot	LLM RAG (OpenAI / Claude) + vector DB	Deferred to post-launch OR rule-based FAQ
CMS	Sanity / Contentful	Markdown in repo or Notion export
Analytics	Mixpanel + GA4	GA4 only

Website pages (sitemap)

1. Home — hero, value prop, AI demo embed, app store badges
2. How it works — 4-step flow (link broker → detect SIP → get dates → track Extra Earnings)
3. AI methodology — model inputs, constraints, audit trail, not a black box
4. Performance / Extra Earnings — backtest results, disclaimers, PaRRVA
5. Pricing — monthly, quarterly, half-yearly
6. Blog — SIP timing education (SEO)
7. Compliance — IA cert, MITC, privacy (DPDP), grievance
8. Support — tickets, SCORES, SMART ODR
9. Install — deep links to App Store / Play Store

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4. Mobile app requirements

Flutter app (Android + iOS) — primary product. Same core features in both scenarios; differences are backend capacity and broker coverage.

ID	FEATURE	REQUIREMENTS
F1	Sign-up & KYC	OTP → PAN → KRA/CKYC → DigiLocker → eSign; HUF support; ≤ 4 screens
F2	Broker linking	OAuth; multi-broker; revoke anytime
F3	MF/ETF import	AA + broker API; filter non-MF/ETF; SIP detection; daily/weekly refresh
F4	Smart-date engine	1–2 dates/fund/month; logged with model version
F5	Order execution	One-tap where supported; else reminder + deep link
F6	Old SIP handling	Guided pause; double-buy alerts
F7	Notifications	Push, SMS (limited), email; category mute rules
F8	Extra Earnings	Shadow ledger vs baseline SIP date
F9	True net returns	XIRR after all costs and estimated tax
F10	Subscriptions	Monthly / quarterly / half-yearly; UPI Autopay; GST invoice
F11	Support	In-app tickets; SCORES link
F12	Hindi + large text	P1 — 200% font scaling

App stack (both scenarios): Flutter · Riverpod/Bloc · FCM/APNs · certificate pinning · Firebase Crashlytics

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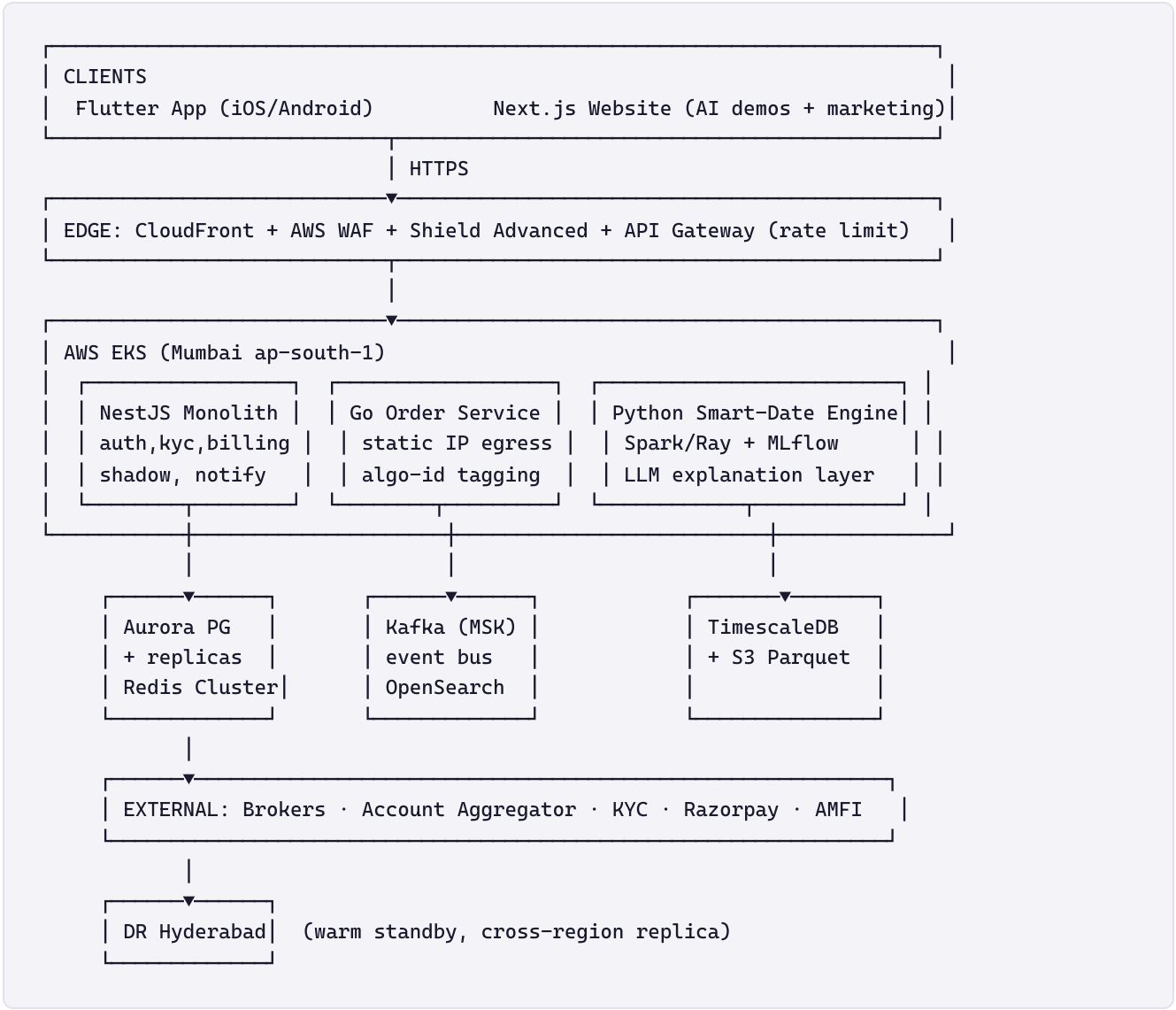
5. Scenario A — Standard architecture NO COST OPTIMIZATION

Built for millions of users from day one. Higher fixed cost, lower rework later.

Stack summary

LAYER	TECHNOLOGY	RATIONALE
Mobile	Flutter	Cross-platform at scale
Website	Next.js 15 + Sanity CMS + CloudFront	Live AI demos, SEO, ISR
API	NestJS modular monolith → microservices path	TypeScript velocity + clean modules
Order service	Go microservice (extracted early)	<3s p95 order latency at scale
AI engine	Python + Spark/Ray batch + MLflow	Millions of fund-month scores/month
LLM layer	GPT-4o-mini / Claude Haiku for explanations	"Why this date" in app + website
Primary DB	Aurora PostgreSQL Multi-AZ + 3 read replicas	99.95% market-hours availability
Time-series	TimescaleDB	NAV history at billion-row scale
Cache	Redis Cluster (ElastiCache)	Sessions, rate limits, schedule cache
Events	Apache Kafka (MSK)	Order/audit replay for compliance
Search	OpenSearch	5-year audit log search
Compute	AWS EKS (Kubernetes)	Auto-scale to millions of requests
DR	Warm standby — Hyderabad (ap-south-2)	RPO ≤ 15 min, RTO ≤ 4 h
Security	WAF + Shield Advanced + KMS	Fintech DDoS protection
Observability	Datadog + PagerDuty	24/7 on-call ready
Brokers (orders)	Zerodha + Groww + Upstox + Angel One	Full API order placement
AA refresh	Daily for all active users	PRD default
SMS	Full PRD spec (fund-ready + double-buy + renewal)	Maximum reach

Architecture diagram



Scale targets (Scenario A)

- 2–10M registered users · 5–15M orders/month
- 99.95% availability during market hours (9 AM – 3:30 PM IST)
- Order p95 ≤ 3 seconds · reminders ±2 minutes

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6. Scenario B — Cost-optimized architecture

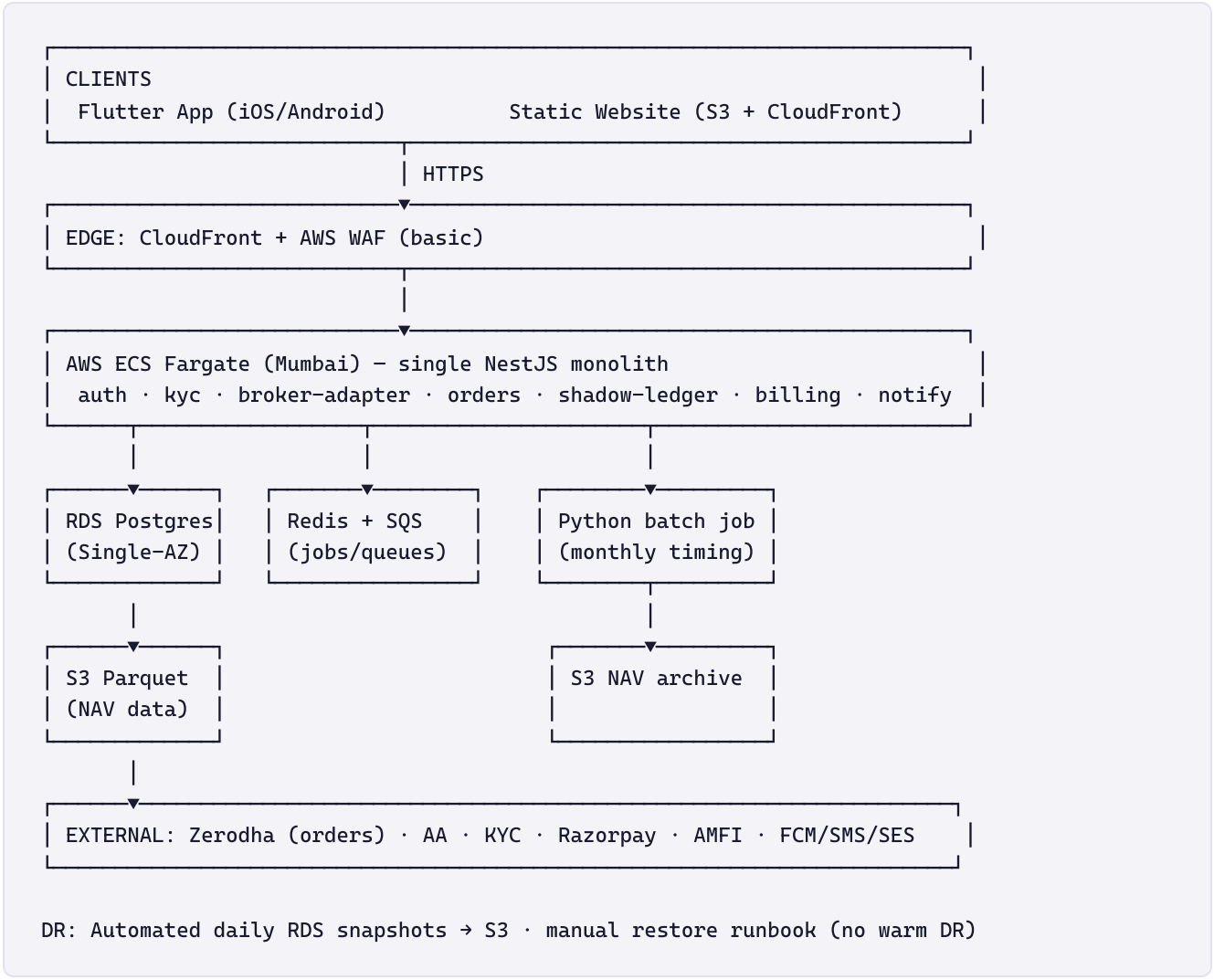
RECOMMENDED FOR LAUNCH

Lean stack to reach Dec 2026 launch and 50k–200k users without over-spending. Scale triggers defined — upgrade only when metrics force it.

Stack summary

LAYER	TECHNOLOGY	WHAT WE CUT VS SCENARIO A
Mobile	Flutter (same)	—
Website	Astro / Next.js SSG on S3 + CloudFront	No live AI API; pre-computed demo data
API	Single NestJS monolith on ECS Fargate	No microservices, no Go order service
AI engine	Python cron on Fargate Spot	No Spark, no GPU, no LLM at v1
Primary DB	RDS PostgreSQL Single-AZ → Multi-AZ at launch	1 read replica only when >20k users
Time-series	S3 Parquet + Athena (NAV history)	No TimescaleDB until needed
Cache	Single Redis node (ElastiCache small)	No Redis Cluster
Events	SQS + EventBridge	No Kafka until 500k users
Search	PostgreSQL full-text (audit logs)	No OpenSearch until compliance search needed
Compute	ECS Fargate (2–4 tasks)	No Kubernetes
DR	Daily snapshots + documented restore runbook	No warm Hyderabad until MRR covers it
Security	WAF + KMS (no Shield Advanced)	Add Shield when traffic justifies
Observability	CloudWatch + Sentry free tier	No Datadog until on-call team exists
Brokers (orders)	Zerodha only at launch	Others: reminder + deep link only
AA refresh	Daily active subscribers; weekly free/lapsed	40–60% AA cost reduction
SMS	Fund-ready + double-buy only (PRD minimum)	No SMS for subscription renewal
FAQ bot	Static FAQ page	No LLM RAG at launch

Architecture diagram



Scale-up triggers (when to migrate toward Scenario A)

TRIGGER	ACTION
>500k users OR order p95 > 3s	Extract Go order service; add read replicas
>1M API requests/day sustained	Move ECS → EKS; horizontal pod autoscaling
AA import backlog > 4 hours	Dedicated import worker fleet
Audit log search complaints from compliance	Add OpenSearch
Monthly timing batch > 1 hour	Add Spark on EMR
Paid MRR > ₹50L/month	Enable warm DR in Hyderabad
Second broker order API ready	Add broker adapter + static IP pool entry

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7. Side-by-side comparison

SCENARIO A

Standard — scale-first

- EKS + Kafka + Aurora + TimescaleDB
- 4 broker order APIs
- LLM explanations + live AI website demos
- Warm DR Hyderabad
- Datadog + Shield Advanced
- Daily AA for all users
- Best for: funded team, aggressive growth, millions target from year 1

SCENARIO B

Cost-optimized — launch-first

- ECS + SQS + RDS + S3 Parquet
- Zerodha orders only; others deep link
- Static AI demos; no LLM at v1
- Backup-only DR
- CloudWatch + Sentry
- Tiered AA refresh
- Best for: Dec 2026 launch, 50k–200k year 1, optimize burn

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8. Cost breakdown

All figures in INR (₹), approximate. Vendor costs dominate at every scale — not AWS.

8.1 Monthly run-rate at 50,000 users (Year 1 PRD target)

COST CATEGORY	SCENARIO A (STANDARD)	SCENARIO B (OPTIMIZED)	NOTES
AWS compute + networking	₹1,10,000	₹35,000	EKS vs ECS Fargate
Database + cache	₹85,000	₹28,000	Aurora+replicas vs single RDS
DR + security extras	₹45,000	₹5,000	Warm DR vs snapshots only
AI / ML infra	₹38,000	₹18,000	Spark+LLM vs batch only
Observability	₹35,000	₹3,000	Datadog vs CloudWatch
Website hosting + CMS	₹18,000	₹4,000	Vercel+Sanity vs S3 static
Subtotal — infra + AI	**₹3,31,000**	**₹93,000**	
KYC (amortized)	₹1,80,000	₹1,80,000	Same — one-time per user
Account Aggregator	₹95,000	₹45,000	Daily all vs tiered refresh
SMS + push	₹55,000	₹30,000	Full SMS vs minimum PRD
Payment gateway (~2% rev)	₹35,000	₹35,000	Scales with subscription revenue
eSign + DigiLocker + broker API	₹65,000	₹40,000	Multi-broker vs Zerodha only
VAPT + compliance tooling	₹15,000	₹15,000	Non-negotiable
Subtotal — vendors	**₹4,45,000**	**₹3,45,000**	
TOTAL monthly @ 50k users	**₹7,76,000**	**₹4,38,000**	**B saves ~44%**

8.2 Monthly run-rate by stage (Scenario B — recommended)

STAGE	USERS	INFRA + AI	VENDORS	TOTAL / MONTH
Closed beta	200	₹25,000	₹15,000	₹40,000
Launch + 3 months	10,000	₹55,000	₹70,000	₹1,25,000
Year 1 target	50,000	₹93,000	₹3,45,000	₹4,38,000
Growth phase	200,000	₹2,20,000	₹12,00,000	₹14,20,000

8.3 One-time costs to launch (both scenarios)

ITEM	SCENARIO A	SCENARIO B
Engineering (mobile + backend + ML + web)	₹2.0 – 2.5 Cr	₹1.2 – 1.8 Cr
SEBI IA registration + BSE deposit	₹25 – 40 L	₹25 – 40 L
Legal / compliance	₹20 – 30 L	₹15 – 25 L
VAPT + audit	₹12 – 15 L	₹8 – 12 L
UX / brand / website design	₹15 – 20 L	₹8 – 12 L
Total one-time	**₹2.7 – 3.5 Cr**	**₹1.8 – 2.7 Cr**

8.4 AI cost detail (smart-date engine only)

COMPONENT	SCENARIO A	SCENARIO B
Monthly batch compute (50k users × ~5 funds)	₹12,000	₹8,000
NAV data ingest + storage	₹8,000	₹4,000
Model retraining (quarterly)	₹8,000	₹3,000
MLflow / experiment tracking	₹4,000	₹1,000
LLM "why this date" explanations	₹6,000	₹0 (deferred)
Website AI demo API	₹5,000	₹0 (static data)
Per-user AI cost @ 50k	~₹0.86/user/mo	~₹0.32/user/mo

> **Largest cost levers (both scenarios)**
>
> Account Aggregator refresh frequency · SMS volume · number of broker order integrations · KYC volume at signup. Optimizing these saves more than downsizing servers.

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9. Delivery timeline

PHASE	DATES	SCENARIO A DELIVERABLES	SCENARIO B DELIVERABLES
Foundation	Oct 2026	Full AWS landing zone, EKS, Kafka POC, 4 broker contracts	ECS setup, Zerodha + AA + KYC contracts, static website wireframes
Build	Nov 2026	All P0 features + live AI website + LLM layer + multi-broker orders	P0 features + static AI demo site + Zerodha orders only
Beta	Dec 1–20	200 users, full stack VAPT	200 users, lean stack VAPT
Launch	Dec 21–31	Public release, warm DR live	Public release, backup DR runbook

PHASE	DATES	SCENARIO A DELIVERABLES	SCENARIO B DELIVERABLES
Post-launch	Jan 2027+	Scale tuning, family view, auto-execute	Add LLM, 2nd broker orders, read replicas, then EKS/Kafka as triggers hit

> **Realistic note**
>
> PRD allocates only ~2 months for core build (Nov 2026). Scenario B is designed to be achievable in that window. Scenario A realistically needs 6–9 months of engineering before beta.

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10. Compliance & non-negotiables (both scenarios)

These items must not be cut in either scenario — regulatory and reputational risk exceeds any savings.

AREA	REQUIREMENT	CUT IN SCENARIO B?
SEBI IA registration	Non-individual IA before charging for recommendations	No
Advisory audit trail	Every recommendation logged 5+ years with model version	No
Broker token security	KMS encryption; tokens never on device	No
DPDP / data in India	Consent, privacy notice, deletion on closure	No
Order kill switch	Platform-wide halt capability	No
VAPT before launch	Third-party penetration test	No
Shadow ledger (Extra Earnings)	Core product differentiator	No
PaRRVA (if marketing performance)	Verified past performance figures	No
Warm DR Hyderabad	RPO ≤ 15 min	**Yes — defer**
LLM explanations	User-facing AI copy	**Yes — defer**
Multi-broker order APIs	One-tap on all brokers	**Yes — Zerodha first**
Live AI website demos	Real-time API on marketing site	**Yes — static backtests**

Recommended path

> **Launch with Scenario B. Migrate toward Scenario A using the scale-up triggers in Section 6.**
>
> This matches the returns.trade playbook: prove the regulated AI category on one broker and one asset class (SIP/MF), then expand brokers and automation once unit economics work.

Mod Alpha Technical Documentation · Generated 27 September 2026

Based on Mod Alpha Product Requirements Document (21 September 2026) · Benchmark: returns.trade

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